

SHOCK & PRESSORS — Clinical One-Pager

Four categories · hemodynamic profile → treatment lever · 2021 Surviving Sepsis Campaign · IM resident reference

1 · CATEGORIES & HEMODYNAMICS

Type	CVP	CO	SVR	1st lever
Hypovolemic	↓↓	↓	↑	Volume / blood
Cardiogenic	↑	↓↓	↑	Inotrope ± diurese
Obstructive	↑	↓	↑	Relieve cause
Distributive	↓-nml	nml-↑	↓	IVF + NE

Causes: hypovolemic = hemorrhage, GI/renal loss; cardiogenic = HF, valve, arrhythmia; obstructive = PE, tension PTX, tamponade; distributive = sepsis, anaphylaxis, neurogenic, adrenal crisis.

2 · SORT AT THE BEDSIDE FIRST

- **Warm + wide pulse pressure** = low SVR = **distributive**
- **Cold/clammy + narrow pulse pressure** = low output = **cardiogenic or hypovolemic**
- **SvO₂:** ↓ cardiogenic, ↑ established sepsis (↓ early sepsis). Confounded by CO, SaO₂, Hgb.
- POCUS/TTE: EF, IVC fill, effusion, RV strain.

3 · TANK / PUMP / SQUEEZE

TANK (hypovolemic): blood + IVF, 2 large-bore IV. **No pressors until preload repleted.**

PUMP (cardiogenic): dobutamine (mild ↓SVR can ↑BP via ↑CO) ± diuresis. **No bolus IVF in pulmonary edema.**

SQUEEZE (distributive): IVF + norepinephrine to MAP ≥65; add vasopressin early.

OBSTRUCTIVE: fix the lesion — thrombolyse/anticoag (PE), needle/chest tube (PTX), pericardiocentesis (tamponade).

4 · PRESSORS & INOTROPES

Agent	Receptors	Effect / use
Norepinephrine	α1>β1	↑↑SVR, mild↑CO; septic 1st-line
Vasopressin	V1	↑SVR; NE-sparing add-on
Epinephrine	α1,β1,β2	↑HR/CO/SVR; anaphylaxis , refractory
Phenylephrine	α1	↑SVR, reflex↓HR; when tachyarrhythmia limits NE
Dobutamine	β1>β2	↑↑CO, mild↓SVR; cardiogenic
Dopamine	dose-dep	More arrhythmia; largely supplanted by NE

Receptors: α1 vasoconstrict · β1 ↑HR/contractility · β2 vasodilate · V1 non-adrenergic vasoconstrict.

5 · 2021 SURVIVING SEPSIS UPDATES

- **Norepinephrine** remains first-line for septic shock.
- **Balanced crystalloids** (LR/Plasma-Lyte) over 0.9% saline; 30 mL/kg bolus now a **weak** rec.
- **Add vasopressin** (0.03 U/min) at NE ~0.25–0.5 mcg/kg/min rather than escalating NE.
- **Start NE peripherally** to restore MAP — don't wait for a central line.
- **CLOVERS (2023):** restrictive vs liberal fluids = no mortality difference.
- **Angiotensin II** (ATHOS-3): add-on for catecholamine + vasopressin-refractory vasodilatory shock.

DO / AVOID

AVOID: pressors in an unfilled tank (hypovolemia); IVF bolus in cardiogenic pulmonary edema; routine dopamine first-line (arrhythmia); β-blockade in catecholamine excess.

DO: read the exam first (warm/cold, full/empty); MAP goal ≥65; cultures + antibiotics within 1 h and source control in sepsis; epinephrine first for anaphylaxis; a-line for any vasopressor drip.